COER UNIVERSITY, ROORKEE

Department of Computer Science & Engineering

Full Stack Development — Mini E-Commerce Project

Lab Sheet 2: Making It Interactive (Vanilla JavaScript) — Solution

Covers Project Phase 3

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Course	B.Tech CSE — 5th Semester
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Repository	https://github.com/cu24250214-dotcom/Full_stack_BTCS303P
Objective	Add cart logic, dynamic rendering, and form validation using plain JavaScript before moving to React.
Duration	2 hours
Prerequisite	Lab Sheet 1 completed (static + styled pages)
Submission	script.js linked to relevant pages, plus updated HTML if needed

Tasks

1. Create a script.js file. In it, define a JavaScript array of product objects (id, name, price, image) representing your store's catalog — at least 8 products.
2. On products.html, use JavaScript to dynamically render the product cards from your array into the page (instead of hardcoded HTML blocks) using DOM methods like createElement / innerHTML with template literals.
3. Implement an 'Add to Cart' click handler on each product card that adds the product to a cart array (store it in a JS variable, and also persist it using localStorage so it survives a page refresh).
4. On cart.html, read the cart from localStorage and dynamically render each item with a working quantity input and a 'Remove' button.
5. Write a function calculateTotal() that recalculates and displays the cart's grand total whenever quantity changes or an item is removed.
6. On checkout.html, add form validation using JavaScript: Name and Address must not be empty, Pincode must be exactly 6 digits, Phone must be exactly 10 digits. Show inline error messages instead of using alert().
7. On successful validation, show a simple confirmation message on the page (e.g., 'Order placed!') and clear the cart from localStorage.
8. Add a small script on every page to update a 'cart item count' badge next to the Cart link in the navbar, read from localStorage.

Concepts You Must Know

• Array methods: map(), filter(), find(), reduce() (for total calculation)

• DOM manipulation: querySelector, addEventListener, createElement

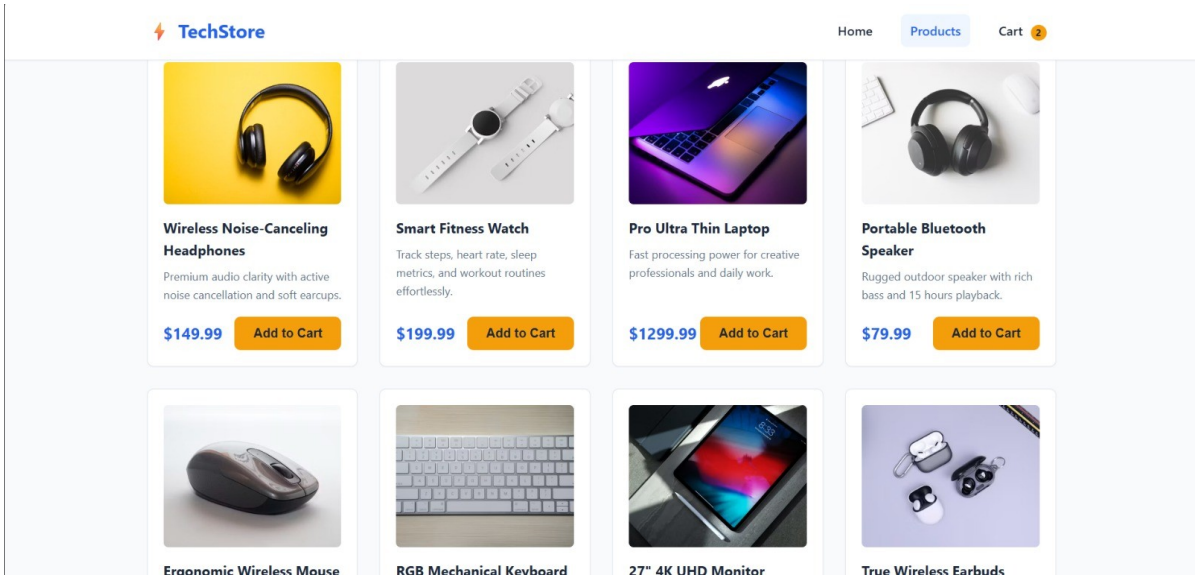
• localStorage.setItem() / getItem() / JSON.stringify() / JSON.parse()

• Basic form validation with regular expressions or simple length/pattern checks

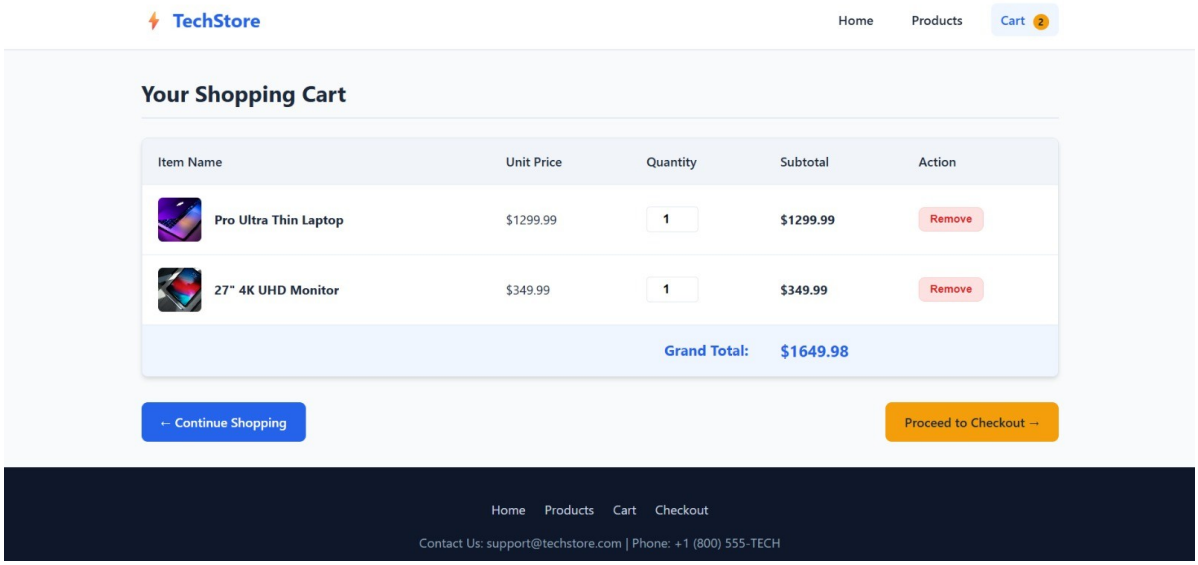
Solution — Output Screenshots

The following screenshots demonstrate the completed interactive storefront, showing dynamic product rendering, cart logic, and checkout form validation implemented using vanilla JavaScript.

1. Products Page — Dynamic Rendering & Add to Cart



2. Cart Page — Quantity, Remove & Grand Total



3. Checkout Page — Inline Form Validation

